When did living land plants arise? What their fossil age limits require

Barry W. Brook*[1], Claude Opus 5.5 (Anthropic) and OpenAI Codex[2]

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1. University of Tasmania, Hobart, Tasmania, Australia
2. AI research collaborators; roles are stated in the Declaration of AI use

Corresponding author: Barry W. Brook

Abstract

Molecular clocks date the ancestor of living land plants from the Cambrian to the Late Ordovician, partly through different fossil maximum ages. We examined three maxima: 515.5 and 459.1 million years ago (Ma) for living land plants, and 431.2 Ma for living vascular plants (clubmosses, ferns and seed plants). We traced their sources, modelled Libyan spore records and set a limit on when the vascular-plant body-fossil record began. The oldest rests on a qualitative preservation argument. Libyan boreholes yield three-rayed (trilete) spores in 51 of 59 early Silurian samples and none of 28 older ones. If spores reached the rocks equally easily throughout, steady supply since 459.1 Ma is limited to 9% of the later supply (95% limit). If vascular-plant fossils were found at a steady rate, the 95% confidence limit on their record's start is 428.8 Ma; counting fewer independent finds gives 432.4–434.05 Ma. Ancestors at 444–452 Ma fit if fossils were found 8–12 times less often before the oldest find than after it (2.4–4.5 with fewer finds), with recognisable bodies appearing with the ancestor. Older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

Keywords: molecular clock; fossil calibration; land plants; vascular plants; stratigraphic confidence interval; Ordovician–Silurian

Introduction

Molecular clocks estimate when lineages diverged by combining DNA differences with fossil limits on the ages of selected ancestors. A crown group comprises the most recent ancestor shared by its living members and all its descendants, including extinct lineages. For the ancestor of all living land plants (crown embryophytes), Morris et al. [1] obtained 515–474 Ma across seven tree topologies and Harris et al. [2] 515–493 Ma. Hooper et al. [3] reanalysed Harris et al.'s DNA data with revised fossil calibrations and obtained 457–450 Ma, a mean 55 Myr younger. For the ancestor of living vascular plants (crown tracheophytes: clubmosses and their relatives, ferns and seed plants), the three studies give 451–419, 452–447 and 432–428 Ma (figure 1); Su et al. [4] obtained older land-plant ages still, under a Proterozoic maximum. The dates bear on the order in which plant characters were assembled, and on how early plants changed productivity, weathering and climate [5–8]. They date the ancestors of living plants; extinct lineages may have reached land earlier.

A fossil sets a minimum age: its lineage existed by the age of its oldest secure member. A maximum based on absence adds an argument that older rocks would probably have preserved and yielded recognisable remains had the lineage existed [9]. How calibrations are specified can strongly shape estimated dates, and with large sequence datasets they can dominate the uncertainty in absolute ages [10,11]. A soft maximum permits older ages through a specified tail of the calibration prior [12]. Whether absence can set a maximum is contested [13,14], and best practice asks each calibration to justify its maximum from the relevant geological, geographical and taphonomic record [15].

Linking a fossil to an ancestor crosses two intervals: from divergence to the first recognisable character, and from that character to its first recorded fossil [16]. Both are uncertain for early land plants. Dispersed spores appear tens of millions of years before branched plant bodies [17–19], terrestrial deposits are scarce before the latest Silurian, at least in western Europe [5], and the oldest parent plants known in detail are minute [20].

Three maxima divide the current clocks (table 1). Morris et al. set 515.5 Ma for living land plants, arguing that older marine rocks rich in algal cysts would also have yielded land-plant spores [1,14]. Hooper et al. set 459.1 Ma from the oldest possible trilete (three-rayed) spores, in the Borenshult-1 core of southern Sweden [21], and 431.2 Ma for living vascular plants from Irish *Cooksonia* [22]. The three therefore rest on a preservation analogy, on dispersed spores and on plant bodies, each with its own route into the record, and each needs its own test.

We ask three questions of the published record. What evidence supports each maximum, and what links connect it to the dated ancestor? How much steady spore supply before the Silurian do the records of one Libyan basin allow? How much less often would early vascular-plant fossils have to be found, or how long after the ancestor would recognisable bodies have to appear, for older dates to fit? Established theory sets an expectation for each. An argument from absence needs documented chances for remains to be delivered, preserved and found [15,16]. Where older and younger rocks were sampled comparably, each further sample reporting no target spores tightens the limit on a high, steady spore supply. With the oldest find and the time window fixed, more independent finds bring the limit on a record's start closer to that find, and a lower finding rate before it pushes the limit back [23,24]. The audit identifies what each maximum assumes; the models limit local spore supply and describe the finding rates or delays that would fit older vascular-plant dates.

Table 1. The three maxima examined: the evidence each rests on, the test the published record allows, and the links a clock must still assume. GTS2012, Geologic Time Scale 2012.

Maximum	Set by	Evidence	Test available	Links assumed
515.5 Ma, living land plants	Morris et al.	Base of the *Olenellus* Biozone, below the oldest possible Cambrian spore-like packets; older marine rocks with algal cysts	Affinity of the Cambrian packets; any fully reported older sequence	Spore delivery to marine basins, catchment, search effort, spore production by the earliest members

Maximum	Set by	Evidence	Test available	Links assumed
459.1 Ma, living land plants	Hooper et al.	GTS2012 Sandbian base plus uncertainty, keyed to the oldest possible trilete spore (Borenshult-1, Sweden)	Older samples reporting no triletes against younger samples reporting them, Murzuq Basin, Libya	Earliest crown members produced triletes, the trait persisted, producers lived in the catchment
431.2 Ma, living vascular plants	Hooper et al.	GTS2012 Homerian base plus uncertainty, keyed to Irish Homerian *Cooksonia*	A limit on when the body-fossil record began, and what older ancestors require	Finding rate before the oldest find, delay before recognisable bodies, the oldest finds lie within the vascular crown

Material and methods

(a) Timescale and interpretations. Analysed ages use the stage bases of the International Chronostratigraphic Chart, v2024/12 [25]; literature ages were mapped by stage. Published maxima keep their source timescale, GTS2012, so they can be compared as their authors set them. Each analysis declares which interpretation of disputed identifications it uses, main or inclusive, so contested fossils enter as alternatives.

(b) Source audit. For each maximum we traced the calibration fossil, its age rule and how it was found back to the original descriptions, and recorded each published interpretation of the fossil's identity with its evidence [15]. A reported absence applies to the material examined, and a missing report counts as an absence where the source reports the target fully.

(c) Libyan spore records. Abuhmida [26] reported spores and other microfossils from Ordovician to lower Silurian wells in the Murzuq Basin, southwest Libya. We transcribed all 164 appendix samples with their chart and prose reports. The analysis uses the chart-coded samples of drill cuttings (rock chips returned during drilling) from five wells: three supply 28 samples older than the Llandovery (earliest Silurian), and all five supply 59 Llandovery samples, which serve as the reference. The source states the absence of trilete spores explicitly for part of the sequence; for the Hawaz Formation, samples reporting none rest on complete reporting charts and the stated microscopy protocol. The basin lay near the Ordovician South Pole, within the area of Hirnantian glaciation [27,28] (figure 3*a*).

(d) Libyan spore model. We modelled whether trilete spores were reported in each sample. Spore supply before the Llandovery is a fraction f of Llandovery supply, and a fixed delivery ratio ρ covers differences between older and younger rocks in transport, preservation and preparation ($\rho = 1$: spores reached both equally easily). Samples from one well share a well effect, gamma-distributed with mean one (shape and rate 2), and are otherwise independent; each sample's age is uniform within its stage range. Fitting the Llandovery rate and f together, we calculated upper 95% limits on f for steady supply from a given start, and for supply that varied between zero and the Llandovery level, averaged with each interval weighted by how much of the older samples' age ranges falls within it. Using the younger reports as the reference, we first checked that the design could exclude steady supply at the Llandovery level if every older sample reported no triletes and fell within the period of supply; each simulated dataset faced the same check. We then tested the method on simulated data with known answers. Among simulated datasets passing that check, the method had either to withhold a limit or to give one at or above the true value; a quantity was used only if this held in more than 93% of trials with 95% confidence, with a preplanned pooled rule for borderline cases (electronic supplementary material, S2).

(e) Body-fossil finds. We assembled land-plant body fossils dated to 453–393 Ma from Paleobiology Database (PBDB) occurrences, whose names an AI model classified with cited sources before our audit, and

from a source-reviewed set of the oldest records with their competing interpretations. Collections, taxa or papers sharing a locality and age count as one find, a proxy for an independent fossil horizon. We declare three nested categories. Land-plant bodies are any accepted macro- or mesofossil. Polysporangiophytes have documented branched axes bearing sporangia (spore capsules), or placement within the vascular crown. Crown vascular plants belong to the lycophyte line (clubmosses and relatives) or the euphyllophytes (ferns and seed plants) [29–32]. Under the main interpretation the three categories hold 78, 71 and 50 finds, of which 76, 69 and 48 are certainly older than the end of the Emsian (393.47 Ma), the younger edge of the time window (electronic supplementary material, S5.1).

(f) Limit on when the record began. Let $w(s)$ be a fixed relative rate at which recognisable fossils of a category are found at age s, and $U(a) = \int_c^a w(s)\,ds$ the cumulative finding rate between a younger cutoff c, here the end of the Emsian, and age a. For n independent finds with oldest age A, the 95% confidence limit T on the start of the recognisable record solves $U(T) = U(A)\,0.05^{-1/n}$ (figure 2). With constant w this is the classical confidence interval on a stratigraphic range, and the absolute rate cancels [23,24]; varying w follows the extensions to unevenly spaced fossil horizons [33,34]. For finds dated by age ranges, A is the oldest possible age of the oldest find, and n counts only finds certainly inside the window; both choices move the limit older. The limit concerns the start of the record; the Discussion treats its link to ancestors.

(g) Finding-rate histories and shape tests. We compared six assumed histories of the finding rate: steady; proportional to PBDB land and marginal-marine collections per Myr [35], with or without the collections that hold plant fossils; declining exponentially into the past with 25 or 10 Myr timescales; and plant-excluded collection density multiplied by the 10 Myr decline. The collection records reach back to 469.4 Ma, the oldest age the analysis can consider. For each history, an exact Kolmogorov–Smirnov test compared the spread of fossil ages it predicts with the observed spread, given the count and the oldest find. Because many ages are known only as ranges, we searched every possible assignment of ages within those ranges, including whether the finds straddling the cutoff fall inside the window, for the best fit [compare 36]. A history that fails under every assignment is rejected (electronic supplementary material, S5.2).

(h) Lower earlier finding rates and delays. The main scenario holds the finding rate steady after 426.7 Ma, the oldest possible age of a crown vascular-plant find, and lowers it before that age. For a candidate ancestor age t and an assumed delay d before the first recognisable bodies, the ratio of earlier to later finding rate that puts $t - d$ where the limit of (f) would fall is $r = (A - c)(0.05^{-1/n} - 1)/(t - d - A)$, defined where $t - d > A$; fossils would then have to be found $1/r$ times less often before the oldest find than after it. Longer delays reduce the reduction a given ancestor age needs [16]. We chose where to put the step after seeing the data, so these curves show what would fit, without the 95% confidence that (f) gives each fixed history.

Results

(a) The 515.5 Ma maximum rests on a qualitative argument. Morris et al. placed this maximum below the oldest possible total-group records, spore-like packets and clusters from the Cambrian Stage 4 Rome Formation, at the base of the *Olenellus* Biozone (about 515.5 Ma). Their preservation argument, from older marine rocks with algal cysts and without land-plant spores, shows that walled microfossils survived and were recovered [1,14]. How many spores reached those basins, the land areas they drained and the search effort remain unmeasured in the sources we assessed. All three sources we examined assign the Cambrian packets to algae or lineages before land plants: Charophyta [37], a pre-embryophytic grade [38] or ancestral aeroterrestrial algae [39] (electronic supplementary material, S4).

(b) The younger maxima follow a stage-base rule. Hooper et al. set both younger maxima as a GTS2012 stage base plus its uncertainty. They keyed 459.1 Ma (458.4 + 0.7 = 459.1 Ma) to a possible trilete spore at 48.2 m in the Borenshult-1 core [21]. The oldest definite trilete level, at 43.1 m, is also Sandbian and lies below the 454.41 Ma Kinnekulle K-bentonite, so the rule gives the same value from either level (electronic supplementary material, S2.4). They keyed 431.2 Ma (430.5 + 0.7 = 431.2 Ma) to Irish Homerian *Cooksonia* [22]. The older Czech *Cooksonia barrandei*, a branched sporophyte with terminal sporangia from a Sheinwoodian (early Silurian) horizon dated from its slab fauna and by correlation [30], meets the same criterion (electronic supplementary material, table S13). On the current timescale the rule gives outer values of 431.9 Ma for the Homerian and 434.1 Ma for the Sheinwoodian.

(c) Older Libyan samples reporting no triletes limit a steady, well-delivered spore supply. Trilete spores are reported in 51 of 59 Llandovery samples and in none of the 28 older samples (figure 3*a*). If spores reached both sets of rocks equally easily, a steady supply since 459.1 Ma is limited to 0.091 of the Llandovery supply, and a supply varying between zero and the Llandovery level to an average of 0.184 (figure 3*b*; electronic supplementary material, table S12). If older spores reached the rocks a tenth as easily, steady supply near the Llandovery level fits (limit 0.910). For steady supply the samples limit the product ρf, which stays at 0.091 whatever the delivery.

(d) The Libyan limits cover mainly the later Ordovician. Most of the older samples' weight comes from 18 Katian–Hirnantian (Late Ordovician) samples, all younger than 459.1 Ma. Only 10 broadly dated Darriwilian (Middle Ordovician) samples, from two wells, have age ranges reaching beyond 459.1 Ma. Their contribution under a start at that age depends on how ages are spread within the stage; placed at stage midpoints, they contribute nothing. The limits therefore concern steady supply through the later Ordovician in one high-latitude basin.

(e) At a steady finding rate, the 95% confidence limit on the vascular-plant body record's start is near 429 Ma. If crown vascular-plant fossils were found at a steady rate, the 48 finds certainly inside the window put the 95% confidence limit on the start of their record at 428.8 Ma (figure 4*a*). Finding rates that track collection density give 428.9 Ma, or 428.5 Ma with plant-bearing collections removed, and a 25 Myr decline into the past gives 431.6 Ma. If finding fossils became steeply harder with age (the 10 Myr decline), the method sets no limit as far back as 469.4 Ma, the edge of our data. That history, however, predicts a spread of fossil ages unlike the one observed, however the uncertain ages are assigned. The 25 Myr decline passes the shape test under one assignment of ages when all 50 possible finds are counted ($p = 0.067$), and fails under every assignment of the 48 finds certainly inside the window (best $p = 0.035$).

(f) Counting fewer independent finds moves the limit past 431.2 Ma. Treating each region and stage as one find (19 finds) gives 432.4 Ma, and counting only the curated oldest records (15 finds) gives 434.05 Ma, both at a steady finding rate. Shifting stage boundaries by their quoted uncertainties, in the scenarios tested, moves the main limit between 427.31 and 430.42 Ma (electronic supplementary material, S5.3).

(g) Older vascular-plant dates fit if early fossils were found less often, or recognisable bodies appeared later. In the main scenario, halving the finding rate before the oldest find moves the limit to 431.0 Ma, still younger than the 431.2 Ma maximum. Ancestors at 444–452 Ma, spanning the Morris et al. mean dates and the Harris et al. interval, fit if fossils were found 8–12 times less often before the oldest find than after it, with recognisable bodies appearing with the ancestor (figure 4*b*). Counting by region and stage, or only the curated oldest records, 2.4–4.5 times suffices, and 431.2 Ma already fits a steady rate. At a steady finding rate, the same ancestors fit if recognisable bodies appeared at least 15–23 Myr after them. Smaller reductions combine with shorter delays (figure 4*b*).

(h) One Ordovician find moves the broader land-plant limit. Sporangial fragments with in situ cryptospores from the Ghaba-1 borehole, Oman, dated to the Katian on later correlation [40], form one land-plant body find. Adding them moves the steady-rate limit for all land-plant body fossils from 434.5 to 455.2 Ma, and the enlarged set fails the shape test under all six histories. The fragments came from palynological preparation of borehole material, whereas most later finds are compressions collected from bedding planes [41].

Discussion

The early record allows older ancestors, and the analyses state what each would require. In the Murzuq Basin, the older samples exclude a steady, well-delivered trilete supply at the Llandovery level through the later Ordovician; lower delivery, lower local supply or spore producers outside the catchment all fit. At a steady finding rate, or one tracking collection density, the 95% confidence limit on the start of the recognisable crown vascular-plant record lies near 429 Ma, and counting fewer independent finds puts it at 432.4–434.05 Ma. For each way of counting, a lower earlier finding rate, later recognisable bodies or both fit ancestors older than these limits; the models give the size of each requirement in aggregate.

Several processes could have made early plants scarcer in the record. Producers may have been rarer, more

geographically restricted, concentrated where little terrestrial sediment accumulated, or preserved in intervals that have been searched less. Body size, cuticle and lignified tissue could also raise the chance that a body is preserved, found and recognised. Branched sporophytes and conducting tissue were acquired stepwise along the lineages being dated [42]. The early branched sporophyte *C. barrandei* is preserved in marine sediments [30], and the Oman fragments show that small sporangia can survive [40]. The models give the combined reduction a candidate age requires; the contribution and plausibility of each process remain open.

Silvestro et al. [43] modelled genus-level PBDB plant occurrences with a joint Bayesian model of preservation and diversification, and estimated the origin of vascular plants at a mean of 433.5 Ma (95% highest posterior density 449.0–424.0 Ma). They estimate when the earliest vascular-plant genus lineage arose, with a fitted, varying preservation rate; we limit the start of the recognisable record of a declared crown category under assumed finding-rate histories, then ask what older ancestors require. The two analyses target different quantities from partly shared data and test different assumptions. CladeDate builds calibration densities directly from fossil times [44], and Hooper et al. link spore-character reconstruction to revised calibrations. We make the maxima's observational requirements explicit, limiting local spore supply and sizing the reductions in finding rate, or the delays, that fit older vascular-plant dates.

Conditions. The spore limits assume that the Llandovery spores are indigenous, that older and younger samples were reported comparably, that ages are uniform within stages, and that samples are independent apart from a shared well effect. Three exploratory checks of that dependence, untested on simulated data, give 0.0995 with a separate Llandovery rate for each well and no well effect, about 0.4 when samples are weighted by well and rock unit, and 0.185–0.365 (varying supply) with one well left out at a time. Linking the local spore record to ancestors requires that the earliest crown embryophytes produced recognisable trilete spores, that the trait persisted and that producers lived in the catchment (electronic supplementary material, S6).

The body-fossil limits and shape rejections assume independent ages on a continuous scale; fixed finding-rate histories, cutoff, timescale, inventory and counting rules; and an observation process whose reported age ranges always contain the true ages, including whether each find falls inside the window. Geological age ranges carry no demonstrated joint coverage; if the probability that any range misses its true age is δ, the test's error rate is at most $\alpha + \delta$, where α is its level for exact ages. Simulations with exact ages gave coverage of 0.941–0.958, falling to about 0.78 when paired duplicate horizons were counted as independent (electronic supplementary material, S5.6). The step scenario was chosen after seeing the data. Linking the vascular-plant limit to ancestors requires that the oldest finds belong to the crown group, and a specified delay before recognisable bodies.

Published evidence could narrow these requirements. Records of searched Silurian terrestrial beds, with effort and empty beds included, would help constrain early finding rates, and compilations of rock units could supply proxies for the terrestrial rock available at each age [45]. A second complete spore sequence outside polar Gondwana, with older samples reporting no triletes, younger samples reporting them and counted target and control grains, would test whether the Libyan result is general; a fully reported marine sequence older than 515.5 Ma, with comparable recovery documented, would help quantify the oldest maximum's absence argument. Finer, independent ages for the Yea (Victoria, Australia) and Bathurst Island (Arctic Canada) horizons, which hold the oldest possible crown vascular-plant finds, would sharpen the body-fossil limit [29,31]. Characters placing the oldest branched sporophytes on the lycophyte or euphyllophyte side of the vascular crown could change which fossils enter the category, as well as their link to the ancestor.

Conclusion

A fossil maximum based on absence combines expectations about ancestral characters with the chance of finding their remains in older rocks. Clock builders can state each maximum's character, geographical and finding-rate assumptions, and justify the probability assigned to older ages against them [12,15]. Testing alternative defensible maxima then shows how much each assumption moves the timescale, and sampling the joint age prior without molecular data, then comparing its effective marginal ages with the posteriors, shows what the molecular data add [46]. The same separation of the evidence, the chance of finding it and its link to the ancestor applies to any maximum argued from absence. Older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

Ethics

We analysed published descriptions and data. The study involved no fieldwork, specimen collection or research involving human participants.

Data accessibility

The repository https://github.com/BWBrook/silurian_plants holds the source-linked derived tables, model code, qualification evidence, body-record outputs with hash receipts, figure and table scripts, and the claims ledger that binds every reported number to its evidence. The frozen PBDB collections extract used by the effort histories is archived under CC0 [35]. Copyrighted full texts are excluded from the repository; source routes and held-file hashes are recorded separately. Openly licensed third-party deposits are fetched by script and hash-checked. Supplementary material is available online.

Declaration of AI use

Claude Opus 5.5 (Anthropic) developed and ran analyses and drafted and revised the manuscript. OpenAI Codex (`gpt-6.1-sol` and `gpt-6.1-terra`) curated and checked source data, implemented analyses, audited statistical methods and reproduction, and revised the manuscript. Earlier Codex work used `gpt-5.6-sol` and `gpt-5.6-luna` for source recovery and methodological checks. Claude and Codex reviewed each other's work. GPT-6 Pro (OpenAI) supplied independent critiques, which were checked against sources and saved outputs before integration. Barry W. Brook originated and guided the programme and contributed scientific decisions; he will review the final manuscript before submission. Task-specific roles, model versions and the limits of sessions whose exact variant was unavailable are recorded in `docs/charter.md`, `docs/audits/` and `manuscript/reconstruction/`, with adopted decisions in `docs/decisions.md`.

Authors' contributions

[Byline and final CRediT wording remain deferred to the submission phase. A contribution inventory is prepared in `manuscript/reconstruction/4-submission-materials.md`.]

Conflict of interest declaration

[Barry to confirm.]

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Fig. 1.

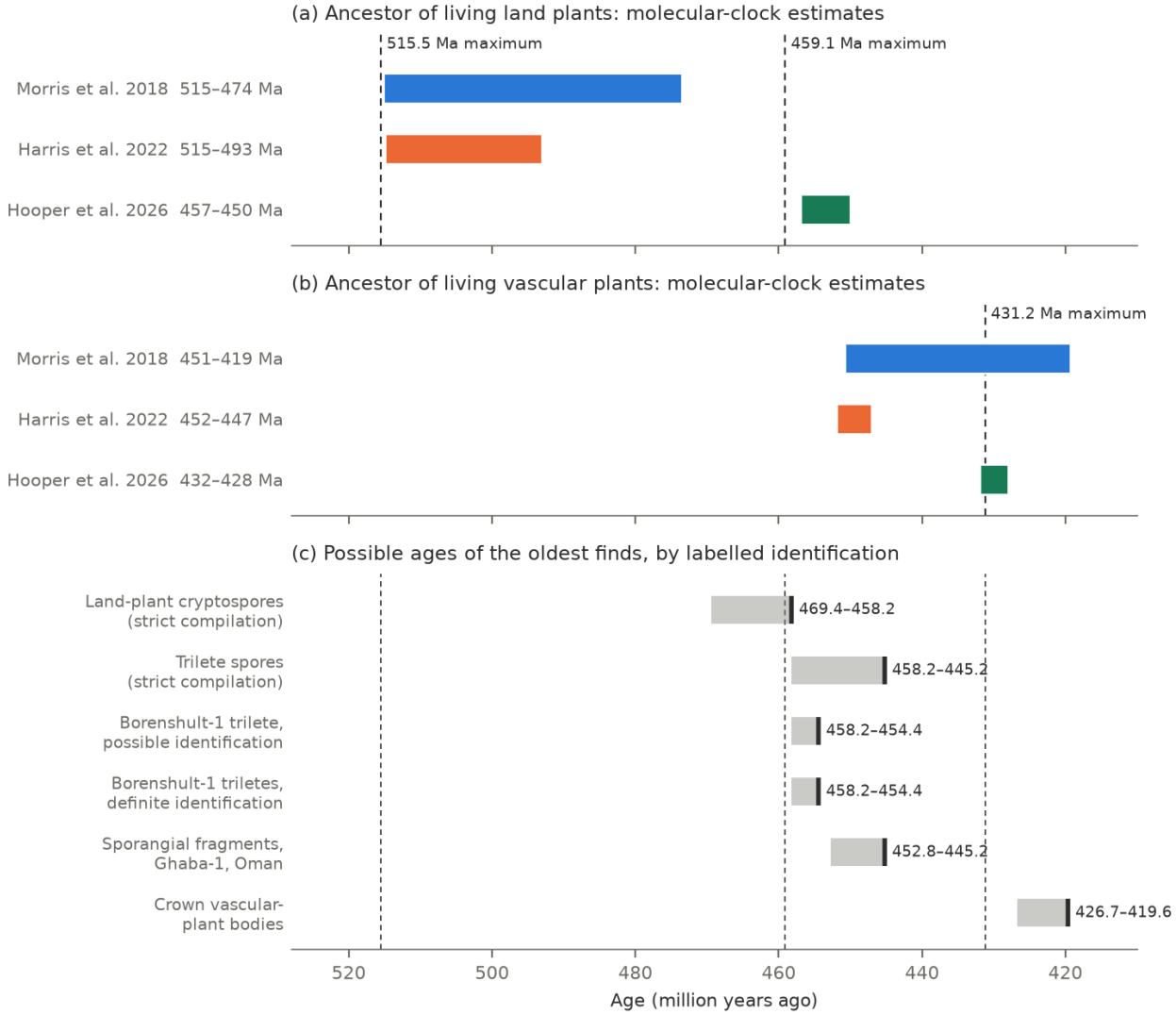


Figure 1: The dating dispute and its fossil evidence. Upper bars give dates for the ancestor of living land plants; the next group gives living vascular plants. Morris et al.'s bars envelope 95% highest posterior density (HPD) intervals across seven tree topologies; Harris et al. and Hooper et al. give their study-specific 95% HPD intervals [1–3]. Vertical lines mark the maxima of 515.5, 459.1 and 431.2 Ma. Clock estimates and maxima keep their published numerical ages. Fossil stage ages use the International Chronostratigraphic Chart v2024/12; independent radiometric ages keep their published values. Each fossil bar spans the possible ages of a category's oldest find, from earliest possible to earliest certain occurrence; the two ends may come from different finds. Separate bars show cryptospores and triletes under the strict compilation interpretation, possible and definite Borenshult triletes [21], Oman's sporangial fragments [40], and crown vascular-plant body fossils [29,31]. The two Borenshult interpretations share this coarse age bracket. Each age holds for the identification labelled; the queried specimen remains a possible trilete.

Fig. 2.

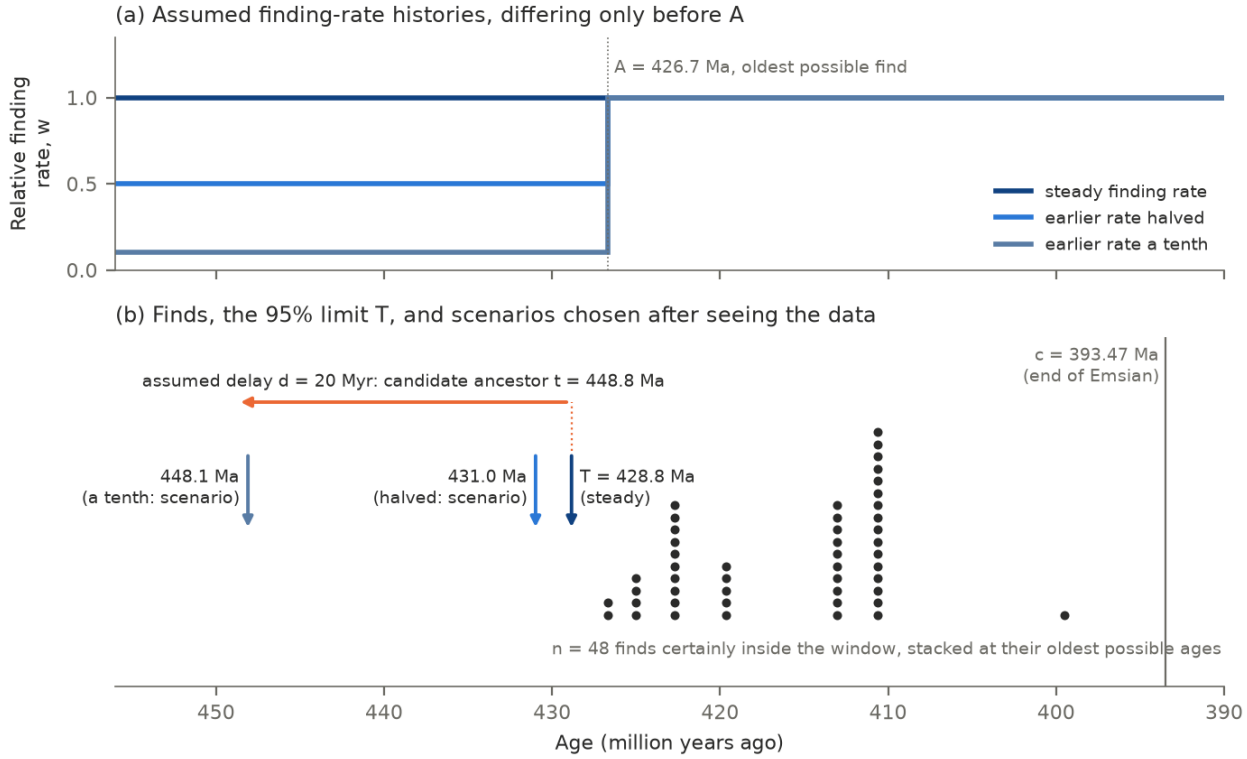


Figure 2: How the body-fossil limit links fossils to ancestors, shown with the main crown vascular-plant finds. (*a*) The finding rate is steady after the oldest find; before it, the rate is unchanged, halved or cut to a tenth. (*b*) Dots show the 48 finds certainly inside the window, at the oldest ends of their age ranges and stacked for visibility. For an assumed finding-rate history, the oldest find A and the count n give the 95% confidence limit T on the start of the recognisable record. For finds dated by age ranges, A is the oldest possible age of the oldest find and n counts only finds certainly inside the window. In the scenarios chosen after seeing the data, halving the earlier finding rate gives a compatibility endpoint of 431.0 Ma, and cutting it to a tenth gives 448.1 Ma; the steady-rate 95% confidence limit is 428.8 Ma. An assumed delay d before recognisable bodies links the start of the record to a candidate ancestor age t; the arrow adds a 20 Myr delay to the steady-rate limit, giving 448.8 Ma.

Fig. 3.

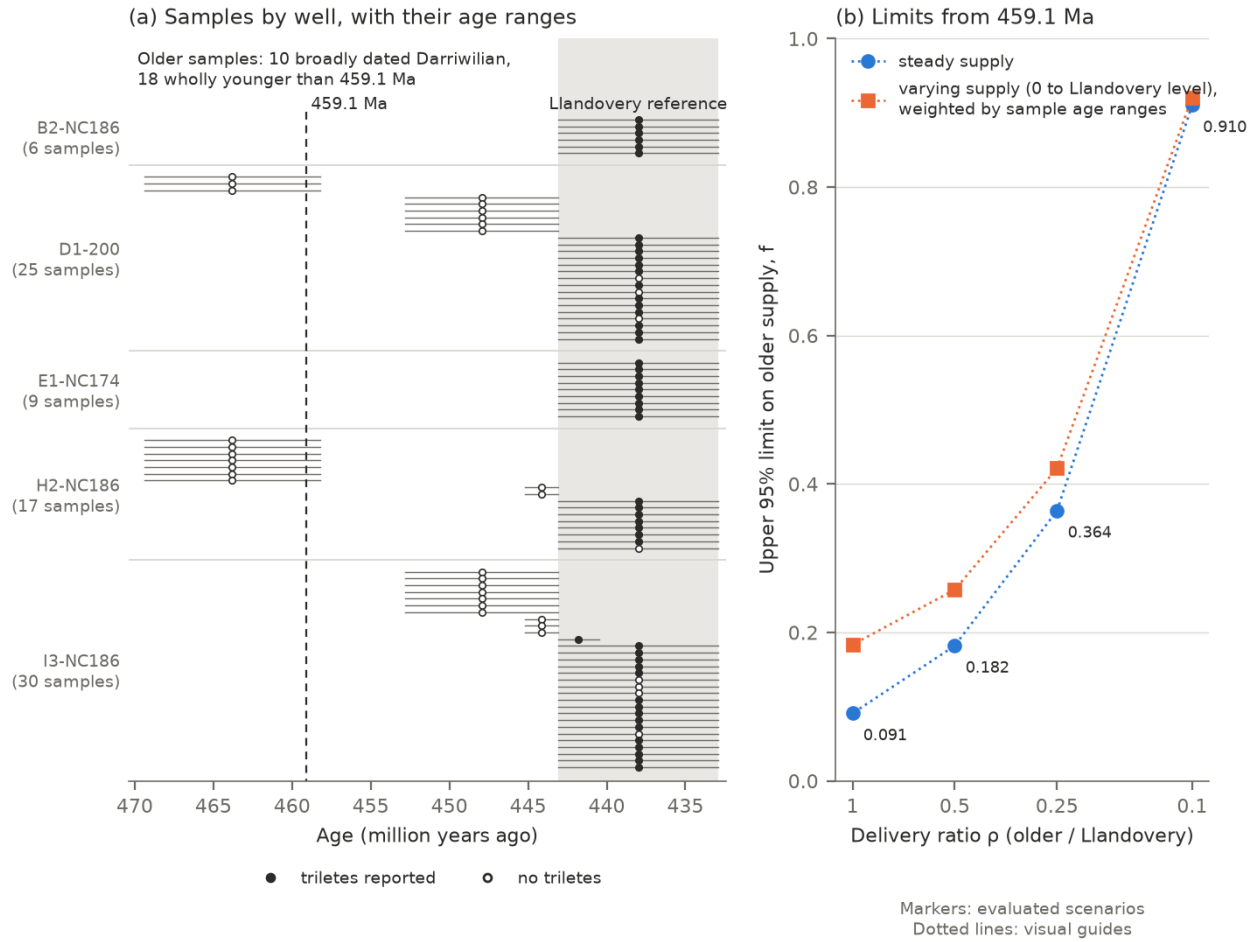


Figure 3: Spore records from the Murzuq Basin, Libya [26]. (*a*) Age ranges of chart-coded samples of drill cuttings (rock chips) from five wells. Filled symbols: trilete spores reported; open symbols: no triletes reported, from explicit statements or complete reporting charts. Symbols sit at the middle of each age range for display; the model spreads each sample's age evenly over its range. Horizontal lines separate wells, with sample counts beside their names. No triletes are reported in any of the 28 older samples; they are reported in 51 of 59 Llandovery (earliest Silurian) samples, the reference. Of the older samples, 18 are wholly younger than 459.1 Ma; only 10 broadly dated Darriwilian samples, in two wells, have age ranges reaching older. Shading marks the Llandovery; the dashed line marks 459.1 Ma. (*b*) Upper 95% limits on older spore supply as a fraction of Llandovery supply, f, for steady supply and for supply varying between zero and the Llandovery level, starting at 459.1 Ma; the varying supply is averaged with each interval weighted by how much of the older samples' age ranges falls within it. The delivery ratio ρ (older to Llandovery) covers transport, preservation and preparation; $\rho = 1$ means spores reached both equally easily. Markers show the four delivery scenarios evaluated; dotted lines are visual guides. At a marked ratio, supply below its limit fits the shared-well model. With poorer delivery, supply near the Llandovery level fits.

Fig. 4.

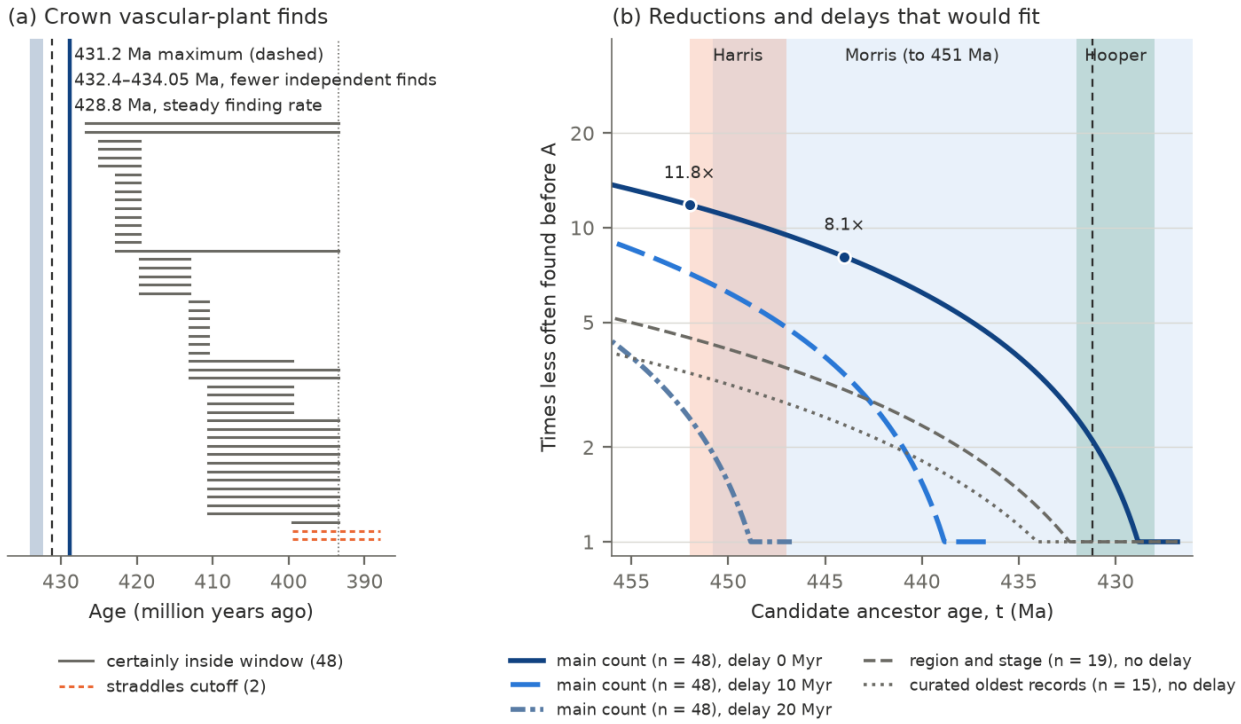


Figure 4: What older vascular-plant ancestors require of the fossil record. (*a*) The 50 possible locality–age finds, ordered by oldest possible age; 48 lie certainly inside the window, and the two straddling the cutoff are marked separately. At a steady finding rate the 95% confidence limit is 428.8 Ma; counting fewer independent finds, by region and stage or only the curated oldest records, gives 432.4–434.05 Ma. (*b*) How many times less often fossils would have to have been found before the oldest find than after it, for each candidate ancestor age t. The finding rate is steady after $A = 426.7$ Ma, the oldest possible age of the oldest find, and the younger cutoff is $c = 393.47$ Ma. Where a curve rises above one, an oldest find no older than A has probability 0.05 given the count, the rule behind the steady-rate limit, with the record starting at $t - d$. Main curves use $n = 48$ and assumed delays d of zero, ten and twenty Myr before recognisable bodies; zero-delay comparison curves count by region and stage ($n = 19$) or only the curated oldest records ($n = 15$). Shading marks the published clock intervals; Morris's younger interval continues outside this panel (figure 1). Combinations with $t - d \leq A$ are omitted; flat portions at one show ages a steady finding rate already allows. We placed the step after seeing the data, so the curves show what would fit and carry no 95% confidence. Older ancestors fit if early fossils were found several times less often, or recognisable bodies appeared later.